AP CALCULUS AB – UNIT 4

Contextual Applications of Differentiation

Comprehensive Coloured Review

Name: ____________ **Class:** ____________ **Date:** ____________

Review Scope and Learning Objectives

- Interpret derivatives and higher derivatives with correct units and contextual meaning.
- Analyze position, velocity, acceleration, speed, direction, and turning behavior.
- Model applied rates and solve related-rates problems from geometric or physical constraints.
- Use linearization and differentials for approximation and error analysis.
- Recognize valid indeterminate forms and apply L'Hopital's Rule appropriately.

Website Lesson Map

Lesson	Website title	Review target
4.1	Interpretations of the Derivative	Interpret, model, calculate, and justify applied derivative conclusions.
4.2	Straight Line Motion	Interpret, model, calculate, and justify applied derivative conclusions.
4.3	Other Rates of Change	Interpret, model, calculate, and justify applied derivative conclusions.
4.4	Related Rates	Interpret, model, calculate, and justify applied derivative conclusions.
4.5	Linear Approximation	Interpret, model, calculate, and justify applied derivative conclusions.
4.6	L'Hopital's Rule	Interpret, model, calculate, and justify applied derivative conclusions.

Key Knowledge, Formulas, and Theorems

- In context, $f'(a)$ is the instantaneous change in output units per input unit at input a.
- Motion: $v = s'$, $a = v' = s''$; speed is $|v|$ and direction changes when v changes sign.
- Related rates: write one constraint, differentiate with respect to time, then substitute the instant.
- Linearization: $L(x) = f(a) + f'(a)(x - a)$; differential estimate $\Delta y \approx dy = f'(a)\,dx$.
- L'Hopital applies only after verifying $0/0$ or ∞/∞; differentiate numerator and denominator separately.

Strategy and Precision Checklist

1. Identify the representation and the exact quantity requested before calculating.
2. Write the relevant definition, theorem, or derivative rule before substitution.
3. Preserve exact values in non-calculator work; use radian mode and appropriate rounding in calculator work.
4. For contextual conclusions, include the input, sign, units, and meaning.
5. Justifications must state why a theorem or procedure applies, not merely name it.

Solved Mixed Examples

Solved Example: Example 1 – 4.2

A particle's velocity is $v(t) = (t-1)(t-2)$. On which interval does it move left?

Answer and Reasoning

Result: $(1, 2)$. The particle moves left where velocity is negative, between the two simple zeros.

Solved Example: Example 2 – 4.3

A quantity Q is increasing at a decreasing rate at $t = a$. Which signs are correct?

Answer and Reasoning

Result: $Q'(a) > 0$ and $Q''(a) < 0$. Increasing means positive first derivative; decreasing rate means the first derivative is falling, so the second derivative is negative.

Solved Example: Example 3 – 4.4

A circle's radius increases at 4 cm/s. How fast is its area increasing when $r = 4$ cm?

Answer and Reasoning

Result: 32π cm^2/s. Differentiate $A = \pi r^2$ with respect to time: $A' = 2\pi r r'$.

Solved Example: Example 4 – 4.4

A 10-m ladder has base $x = 8$ m moving away at 2 m/s. Find the rate of the top.

Answer and Reasoning

Result: -2.667 m/s. Use $x^2 + y^2 = 100$ and $2xx' + 2yy' = 0$; the negative sign means downward.

Leveled Mixed Practice

Foundation

Questions in this band integrate the website lessons at an increasingly demanding level.

1. If $C(t)$ is measured in dollars and t in months, what are the units of $C'(6)$?

2. A particle has position $s(t) = 3t^3 - 4t^2 + 3t$. Find its acceleration $a(t)$.

3. A particle's velocity is $v(t) = (t-1)(t-2)$. On which interval does it move left?

4. A quantity Q is increasing at a decreasing rate at $t = a$. Which signs are correct?

Core

Questions in this band integrate the website lessons at an increasingly demanding level.

5. A circle's radius increases at 4 cm/s. How fast is its area increasing when $r = 4$ cm?

6. A 10-m ladder has base $x = 8$ m moving away at 2 m/s. Find the rate of the top.

7. Given $f(3) = 4$ and $f'(3) = 1$, use linearization to approximate $f(\frac{16}{5})$.

8. Use $f(x) = \sqrt{x}$ linearized at $x = 16$ to approximate $\sqrt{16 + 0.01}$.

Advanced

Questions in this band integrate the website lessons at an increasingly demanding level.

9. Before applying L'Hopital's Rule to a quotient limit, which must be verified?

10. Evaluate $\lim_{x \to 0} \frac{e^{4x} - 1}{x}$.

11. Evaluate $\lim_{x \to \infty} \frac{4x^2 + 1}{e^x}$.

12. If $v(t) > 0$ and $a(t) < 0$, which statement is true?

Difficult

Questions in this band integrate the website lessons at an increasingly demanding level.

13. If $C(t)$ is measured in dollars and t in months, what are the units of $C'(6)$?

14. A particle has position $s(t) = 4t^3 - t^2 + t$. Find its acceleration $a(t)$.

15. Use a calculator to estimate $\frac{e^0.34 - 1}{0.34}$. Which value is closest?

16. A sphere has radius $r = 5.8$ cm and $dr/dt = -0.1$ cm/s. Find dV/dt numerically.

Challenge

Questions in this band integrate the website lessons at an increasingly demanding level.

17. At $t = 2.4$, a particle has velocity -1.7 m/s and acceleration -1.3 m/s^2. Which best describes its motion?

18. A particle has $s(t) = t^3 - 2t^2 + t$. Use a calculator to find $v(0.8)$.

19. Using $f(2.05) \approx 0.7178$ and $f'(2.05) \approx 0.4878$, approximate $\ln(2.07)$.

20. Use a calculator to estimate $\dfrac{e^0.44 - 1}{0.44}$. Which value is closest?

AP-Style Mixed Checkpoint

1. Before applying L'Hopital's Rule to a quotient limit, which must be verified?

(A) The quotient is continuous at the target

(B) The numerator and denominator are polynomials

(C) The quotient has form $0/0$ or ∞/∞

(D) The numerator derivative is zero

2. Evaluate $\lim\limits_{x\to 0} \dfrac{e^{5x} - 1}{x}$.

(A) e^5

(B) 0

(C) 5

(D) 1

3. Evaluate $\lim\limits_{x\to\infty} \dfrac{4x^2 + 1}{e^x}$.

(A) The limit does not exist

(B) $+\infty$

(C) 4

(D) 0

4. If $v(t) > 0$ and $a(t) < 0$, which statement is true?

(A) The particle moves left and slows down

(B) The particle moves right and slows down

(C) The particle moves left and speeds up

(D) The particle moves right and speeds up

5. If $C(t)$ is measured in dollars and t in months, what are the units of $C'(6)$?

(A) dollars per month squared

(B) dollars per month

(C) months per dollar

(D) dollars

6. A particle has position $s(t) = 2t^3 - 4t^2 - t$. Find its acceleration $a(t)$.

(A) 12

(B) $6t^2 - 8t - 1$

(C) $t\left(2t^2 - 4t - 1\right)$

(D) $12t - 8$

7. A particle's velocity is $v(t) = (t-3)(t-6)$. On which interval does it move left?

(A) $(-\infty, 3)$

(B) $(3, 6)$

(C) $(-\infty, 3) \cup (6, \infty)$

(D) $(6, \infty)$

8. A quantity Q is increasing at a decreasing rate at $t = a$. Which signs are correct?

(A) $Q'(a) < 0$ and $Q''(a) > 0$

(B) $Q'(a) < 0$ and $Q''(a) < 0$

(C) $Q'(a) > 0$ and $Q''(a) > 0$

(D) $Q'(a) > 0$ and $Q''(a) < 0$

Free-Response Synthesis

FRQ1 – Straight-line motion

A particle has velocity $v(t) = t^2 - 4t + 3$ for $0 \le t \le 5$.

(a) Find the times when the particle changes direction. [3 points]

(b) Determine intervals on which the particle moves right. [2 points]

(c) Find acceleration and determine when speed increases. [3 points]

FRQ2 – Related rates: conical tank

A cone has radius-to-height ratio $r/h = 1/3$. Water enters at $2\ \text{m}^3/\text{min}$.

(a) Write volume as a function of height only. [2 points]

(b) Find dh/dt when $h = 6$. [3 points]

(c) Interpret the sign and units. [1 points]